PV-Finder Integration and Validation in Athena

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Project Description

With the expected increase of instantaneous luminosity with HL-LHC, work has been done to devise new PV reconstruction algorithms to handle the increased pile-up. PV-Finder is a machine-learning based algorithm that achieves higher resolution with little difference in efficiency and false positive rate than the Adaptive Multi-Vertex Finder (AMVF) algorithm (Lei et al., 2025).

Studies have been done on ATLAS simulated data comparing PV-Finder to the AMVF algorithm, but this has not been done within the Athena framework. The proposed work is to integrate PV-Finder into the Athena framework. This implementation would show how PV-Finder could work in production, and allow for a careful timing and performance study within the Athena framework. Finally, with additional time, I would investigate ways to speed up the algorithm while maintaining physics performance.

Deliverables

- An Athena package with the implementation of PV-Finder as an Athena algorithm.
- Validation of the Athena implementation against existing standalone PV-Finder results.
- A timing and performance study of the Athena implementation on Run 2/3 and HL-LHC samples with performance plots, including findings on potential optimizations.
- Full documentation of the work that has been done in the form of comments and README.

Timeline

Start date: 13 July, 2026

- Week 1-2: Familiarize myself with PV-Finder and the codebase, familiarize myself with the Athena framework.
- Week 3-6: Produce the integration of PV-Finder into Athena framework and validate it against existing standalone PV-Finder results.
- Week 7-8: Using this integration, conduct timing and performance study

- Week 9: Write document and produce complete documentation
- Week 10-12: Investigate potential optimizations, editing and finalizing the document as needed.

References

Qi Bin Lei, Rocky B. Garg, Lauren Tompkins, *Deep Learning for Primary Vertex Identification in the ATLAS Experiment*, CERN Document Server (2025), https://cds.cern.ch/record/2942595.